

# Ibrahim Khaliq

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## EDUCATION

University of Richmond (UR), Richmond, VA

May 2028

Bachelor of Science, Computer Science & Mathematics, Minor in Data Science and Statistics

**Relevant Courses:** Data Structures, Discrete Structures, Software Systems and Development, Linear Algebra, Computer Organization, Machine Learning, Fundamentals of Abstract Mathematics, Business Statistics

## COMPUTER SKILLS

**Languages:** Python, Java, C++, JavaScript, SQL, HTML/CSS

**Frameworks/Libraries:** FastAPI, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Streamlit

**Tools/Platforms:** Git/GitHub, Docker, AWS, Jupyter, Android Studio, Health Connect, Linux

**Concepts:** REST APIs, ETL Pipelines, End-to-End ML Pipelines, Data Processing

## WORK AND LEADERSHIP EXPERIENCE

**Undergraduate Research Assistant, SASSEMA (Psychology Study App), UR**

Jan 2026 – Present

- Led Android port of iOS ADHD ecological momentary assessment app in Java; built onboarding flow, day-2+ recovery logic, and dynamic rendering of 20+ questionnaires (200+ fields).
- Engineered background motion sensing (accelerometer + gyroscope, 25–50 Hz, duty-cycled) with AlarmManager-based reminders for reliable in-the-wild participant data capture.
- Built WiFi-gated batched JSONL upload pipeline (10KB payload threshold, multipart/form-data server sync) to eliminate cellular data cost; integrated Android Health Connect via Kotlin to Java interop for 5 passive data types.
- Diagnosed Android 13 Health Connect platform limitation and proposed hybrid Health Connect + SensorManager fix; authored 22-item code review, ran 45-case QA plan, and shipped debug APK.

**Resident Assistant (RA), Residence Life and Housing, UR**

Aug 2025 – Present

- Serve as mentor and first-response contact for 30+ residents, handling conflict resolution, wellness concerns, policy enforcement, and after-hours emergency response during on-call rotations.
- Design and lead educational and community programs with an 8-person RA team while conducting regular resident check-ins to identify concerns and connect students with campus resources.

**Innovation Fellow, Richmond Innovation Fellows (RIF), UR**

Jan 2025 – Present

- Selected as 1 of 8 fellows from a competitive pool of 100 applicants for a program bridging students with Silicon Valley tech leaders including Google, Adobe, OpenAI, Cisco, and Roku.
- Collaborated in cross-functional teams to design and pitch tech-driven entrepreneurial solutions applying frameworks such as Doblin's Ten Types of Innovation and design-thinking sprints.
- Engaged with venture capitalists, founders, and engineers while applying design thinking and data-driven analysis to evaluate real-world startup challenges and refine product-market fit.

**SWE Intern, Express Tax and Accounting, Falls Church, VA**

Aug 2025 – Sep 2025

- Optimized SQL queries and added composite indexes across PostgreSQL/SQL Server tables, cutting financial report generation time by 40% and eliminating timeout errors on month-end close.
- Built Pydantic validation schemas and structured error-handling across financial reporting pipelines, eliminating 200+ monthly duplicate-entry errors flagged in prior audit cycles and cutting reconciliation time.
- Built Python/Pandas scripts to parse, reconcile, and flag mismatches across 5,000+ monthly ledger transactions from 3 source systems, cutting manual entry time by 35% and surfacing errors before quarterly close.
- Deployed an internal ETL pipeline that automated client tax document ingestion, parsing, and normalization into a central database, saving auditors 10+ hours weekly during peak filing seasons.

## PROJECTS

**AI GitHub Repository Assistant** | Python, FastAPI, Docker, Ollama, GitPython

- Built an AI-powered codebase analysis platform using FastAPI that clones GitHub repositories, generates LLM-based summaries, indexes source code with vector embeddings, and answers natural-language questions about repository architecture.
- Designed scalable retrieval infrastructure with hybrid semantic search, parallel indexing, commit-aware caching, REST APIs, and Docker containerization to support efficient analysis of large repositories.

**Real-Time Facial Recognition System** | Python, OpenCV, face\_recognition (dlib), NumPy

- Built a real-time face detection and recognition pipeline in Python using OpenCV and dlib-based facial encodings to identify individuals from a live webcam feed at 15+ FPS on standard hardware.
- Automated timestamped attendance logging to CSV and designed an extensible identity-registration workflow enabling new faces to be added at runtime without code changes or model retraining.

**Real-Time Voice Assistant** | Python, ElevenLabs Conversational AI SDK

- Developed a real-time, voice-enabled AI assistant using the ElevenLabs Conversational AI SDK, driving natural spoken conversations through live microphone/speaker I/O with sub-500ms response latency.
- Engineered dynamic prompt/context injection and callback-based logging of agent responses, interruptions, and user transcripts, with secure credential management via environment variables.